

1. Introduction

The following report examines crypto traders' behavior in relation to the general market sentiment.

We use two datasets:

Trader activity data: historical executions, PnL, volume, leverage

Bitcoin Fear & Greed Index: A market sentiment score between extreme fear and extreme greed

The main objective is to identify whether traders behave differently during fear versus greed phases and whether sentiment provides hidden signals for risk, profitability, and trade volume.

2. Overview of the Dataset

2.1 Historical Trader Dataset

This dataset contains:

Timestamps of trades

Execution prices

Position size

Profit & loss

Leverage used

Side (long/short)

Account ID

This gives a picture of how traders act, how risky they are, and how profitable they become.

2.2 Fear & Greed Index Dataset

This dataset contains:

UNIX timestamp

Daily sentiment score

Category label (Fear / Neutral / Greed)

It reflects overall crypto market sentiment based on volatility, volume, dominance, social media, and trends.

3. Data Cleaning & Preprocessing

Preliminary work before analysis included several steps:

3.1 Timestamp Conversion

The trader dataset contained timestamps in large microsecond/nanosecond format.

Converted using:

```
pd.to_datetime(timestamp, unit='ms')
```

Extracted date only, not time.

3.2 Cleaning Fear & Greed Dataset

UNIX timestamps converted using unit='s'

Extracted daily dates

Ensured no missing sentiment values.

3.3 Merging

Both data sets were joined on the date column so that trader behavior can be compared with sentiment.

If no matching date exists between the two datasets, rows are dropped.

4. Key Metrics Analyzed

4.1 Profitability (closedPnL)

We compared average PnL on days marked:

Fear

Greed

This shows whether profitable opportunities expand when markets become emotional.

4.2 Trading Volume

We summed volume by sentiment category:

Are traders more active during greed phases?

Does fear cause panic-selling or low participation?

4.3 Leverage Usage

We analyzed average leverage across sentiment categories:

High leverage plus greed may point to FOMO.

Low leverage + fear suggests caution

4.4 Long vs Short Behavior

We analyzed whether more traders:

Go long during greed.

Go short during fear.

5. Findings & Insights

5.1 Profitability Trends

Greed days often align with:

Increased trading activity

More aggressive positions

Higher potential gains and losses

Fear days:

Lower participation

Smaller position sizes

Smoother PnL swings

Insight:

Greed leads to volatility → more opportunities but higher risk.

5.2 Volume & Market Sentiment

Events that tend to cause spikes in trading volume include:

Sudden fear drops (panic exit)

Sudden greed surges (FOMO entry)

Insight:

Volume is a very strong signal for sentiment-driven moves.

5.3 Leverage Behaviour

On greed days:

Traders take higher leverage.

More long positions

FEAR DAYS :

Using conservative leverage

More shorts or no trades

Insight:

Sentiment directly influences risk-taking.

5.4 Strategy Implications

Greed + high leverage = high risk → potential liquidations

Fear plus high volatility is a good mix for short-term scalps.

Neutral sentiment = stable consolidation periods

Sentiment can be used as an additional feature for:

Trading models

Risk management

Predicting Volatility

Timing entries/exits

6. Limitations

Datasets might not perfectly overlap: different date ranges.

Sentiment index is a general indicator, not BTC-specific. Trader data may contain a mix of various account strategies. 7. Conclusion Combining behavioral data with market sentiment allows the trader to understand: When the market is fearful or greedy How traders react emotionally How this affects risk, leverage, volume, and PnL Even simple analysis shows that sentiment is a powerful factor that aligns with market behavior. This insight can be used to build smarter trading systems reacting to emotion-driven signals.